Question	Answer	Marks
2(a)(i)	60×1.6 (= 96 N m) **or** 220×0.40 (= 88 N m)	**C1**
	resultant moment = 96 – 88 = 8.0 N m	**A1**
2(a)(ii)	the resultant force (of A and C) is not zero **or** forces (from A and C) are not parallel **or** forces (from A and C) are not equal (magnitude) **or** forces (from A and C) are not opposite (direction) **or** forces (from A and C) act through the same point so (they are) not a couple	**B1**
2(b)	line with a positive gradient from (t_1, 0) to t_2	**B1**
	line with increasing positive gradient from t_1 to t_2	**B1**
2(c)(i)	<u>sum</u> / <u>total</u> momentum before = <u>sum</u> / <u>total</u> momentum after **or** <u>sum</u> / <u>total</u> momentum (of a system of objects) is constant	**M1**
	if no (resultant) external force / for an isolated system	**A1**

Question	Answer	Marks
2(c)(ii)	$(\Delta)p = F(\Delta)t$	C1
	$m = Ft\,/\,(\Delta)v$ $(m =)\ 5500 \times 0.36\,/\,8.5 = 230$ (kg)	A1
	or	
	$F = ma$ **and** $a = (\Delta)v\,/\,(\Delta)t$	(C1)
	$(m =)\ 5500\,/\,(8.5\,/\,0.36) = 230$ (kg)	(A1)
2(c)(iii)	$(\Delta p =)\ 5500 \times 0.36$ (= 1980 N s) **or** $(\Delta p =)\ 230 \times 8.5$ (= 1955 N s)	C1
	$1980 = (2.5 \times 10^3 - 230)\Delta v$ **or** $1955 = (2.5 \times 10^3 - 230)\Delta v$	C1
	$\Delta v = 0.87\ \text{m s}^{-1}$ **or** $0.86\ \text{m s}^{-1}$	A1
	or	
	$a = 5500\,/\,(2.5 \times 10^3 - 230)$ $(= 2.42\ \text{m s}^{-2})$	(C1)
	$\Delta v = 2.42 \times 0.36$	(C1)
	$\Delta v = 0.87\ \text{m s}^{-1}$ **or** $0.86\ \text{m s}^{-1}$	(A1)

Question	Answer	Marks